A CONTINUOUS SURJECTION TRANSPORTS IRREDUCIBLE COMPONENTS TO IRREDUCIBLE COMPONENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the image of an irreducible subset under a continuous map is irreducible. Invoked inside the counted proof (source lines 1173-1175). Makes the images of components irreducible.
- The statement that the preimage of an irreducible subset under a continuous map whose fibres are irreducible is irreducible. Invoked inside the counted proof (source lines 1176-1178). Shows the map on components is injective.

Source reference. Original lemma lemma-image-irreducible-space: sources/stacks-topology/pinned-original-source.tex, source lines 1086–1086. Navigation pointer only: the counted text refers to the Stacks item lemma-image-irreducible-space, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-irreducible: sources/stacks-topology/pinned-original-source.tex, source lines 1103–1103. Navigation pointer only: the counted text refers to the Stacks item lemma-irreducible, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-pick-irreducible-components: sources/stacks-topology/pinned-original-source.tex, source lines 1142–1142. Navigation pointer only: the counted text refers to the Stacks item lemma-pick-irreducible-components, which is an existing labeled result of this same chapter and is not part of this card.

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Lemma 0.1. *Let $f : X \to Y$ be a surjective, continuous map of topological spaces. If X has a finite number, say n, of irreducible components, then Y has $\leq n$ irreducible components.*

Proof. Say $X_1, \ldots, X_n$ are the irreducible components of X. By Lemmas lemma-image-irreducible-space and lemma-irreducible the closure $Y_i \subset Y$ of $f(X_i)$ is irreducible. Since f is surjective, we see that Y is the union of the Y_i. We may choose a minimal subset $I \subset \{1, \ldots, n\}$ such that $Y = \bigcup_{i \in I} Y_i$. Then we may apply Lemma lemma-pick-irreducible-components to see that the Y_i for $i \in I$ are the irreducible components of Y. $\square$